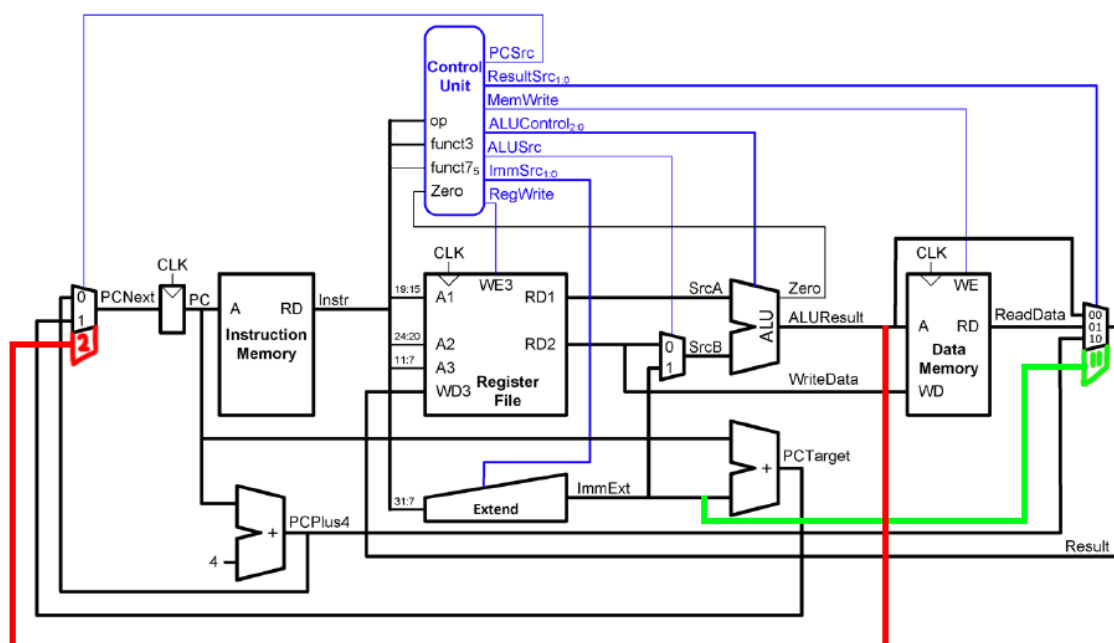


# RISC-V Single-Cycle Data-Path



| type   | instruction | Pc src | Result src | Mem write | Alu control | Alu src | Imm src | Reg write | Alu op | branch | jump |
|--------|-------------|--------|------------|-----------|-------------|---------|---------|-----------|--------|--------|------|
| R-type | Add         | 00     | 00         | 0         | 000         | 0       | X       | 1         | 10     | 0      | 00   |
| R-type | Sub         | 00     | 00         | 0         | 001         | 0       | X       | 1         | 10     | 0      | 00   |
| R-type | And         | 00     | 00         | 0         | 010         | 0       | X       | 1         | 10     | 0      | 00   |
| R-type | Or          | 00     | 00         | 0         | 011         | 0       | X       | 1         | 10     | 0      | 00   |
| R-type | Slt         | 00     | 00         | 0         | 100         | 0       | X       | 1         | 10     | 0      | 00   |
| I-type | Lw          | 00     | 01         | 0         | 000         | 1       | 000     | 1         | 00     | 0      | 00   |
| I-type | Addi        | 00     | 00         | 0         | 001         | 1       | 000     | 1         | 11     | 0      | 00   |
| I-type | Xori        | 00     | 00         | 0         | 110         | 1       | 000     | 1         | 11     | 0      | 00   |
| I-type | Ori         | 00     | 00         | 0         | 011         | 1       | 000     | 1         | 11     | 0      | 00   |
| I-type | Slti        | 00     | 00         | 0         | 100         | 1       | 000     | 1         | 11     | 0      | 00   |
| I-type | Jalr        | 10     | 11         | 0         | 000         | 1       | 000     | 1         | 00     | 0      | 01   |
| B-type | Beq         | 01:00  | X          | 0         | 001         | 0       | 011     | 0         | 01     | 1      | 00   |
| B-type | Bne         | 00:01  | X          | 0         | 001         | 0       | 011     | 0         | 01     | 1      | 00   |
| U-type | Lui         | 00     | 01         | 0         | X           | X       | 100     | 1         | X      | 0      | 00   |
| J-type | jal         | 01     | 11         | 0         | X           | X       | 010     | 1         | X      | 0      | 10   |
| S-type | sw          | 00     | X          | 1         | 000         | 1       | 001     | 0         | 00     | 0      | 00   |

```

int A[10] = {1, 2, 42, 32, 12, 25, 43, 7, -45, 8};
int min = A[0];

for (int i = 1; i < 10; i++)
{
    if (A[i] < min)
    {
        min = A[i];
    }
}

```

```

addi x8, x8, 0x000    -> 0000000000001000000010000010011
addi x5, x0, 13       -> 00000000110100000000001010010011
sw x5, 0(x8)          -> 00000000010101000010000000100011
addi x5, x0, -7        -> 11111110000100000000001010010011
sw x5, 4(x8)          -> 00000000010101000010001000100011
addi x5, x0, 76        -> 00000100110000000000001010010011
sw x5, 8(x8)          -> 00000000010101000010010000100011
addi x5, x0, -20       -> 11111101110000000000001010010011
sw x5, 12(x8)         -> 00000000010101000010011000100011
addi x5, x0, 33        -> 00000010000100000000001010010011
sw x5, 16(x8)         -> 00000000010101000010100000100011
addi x5, x0, 0         -> 00000000000000000000001010010011
sw x5, 20(x8)         -> 00000000010101000010101000100011
addi x5, x0, -11       -> 11111110110100000000001010010011
sw x5, 24(x8)         -> 00000000010101000010110000100011
addi x5, x0, 92        -> 00000101110000000000001010010011
sw x5, 28(x8)         -> 00000000010101000010111000100011
addi x5, x0, -58       -> 11111010001000000000001010010011
sw x5, 32(x8)         -> 00000010010101000010000000100011
addi x5, x0, 19        -> 00000001001100000000001010010011
sw x5, 36(x8)         -> 00000010010101000010001000100011
addi x6, x0, 1         -> 000000000000100000000001100010011
lw x9, 0(x8)          -> 00000000000001000001010000000011
addi x18, x18, 0       -> 00000000000010010000010010010011
addi x5, x5, 40        -> 00000010100000101000001010010011
loop:
beq x18, x5, end_loop  -> 00000000000101001000000001100011
add x19, x18, x8       -> 000000010010010000000010011011011
lw x20, 0(x19)        -> 00000000000010011010010100000011
slt x21, x20, x9       -> 000000010010101001000101011100011
beq x21, x6, end_if    -> 00000000001101011000000100100011
addi x9, x20, 0        -> 00000000000010100000010010010011
end_if:
addi x18, x18, 4       -> 00000000010010010000010010010011
jal loop              -> 1111111010101111111111111011111
end_loop:

```

